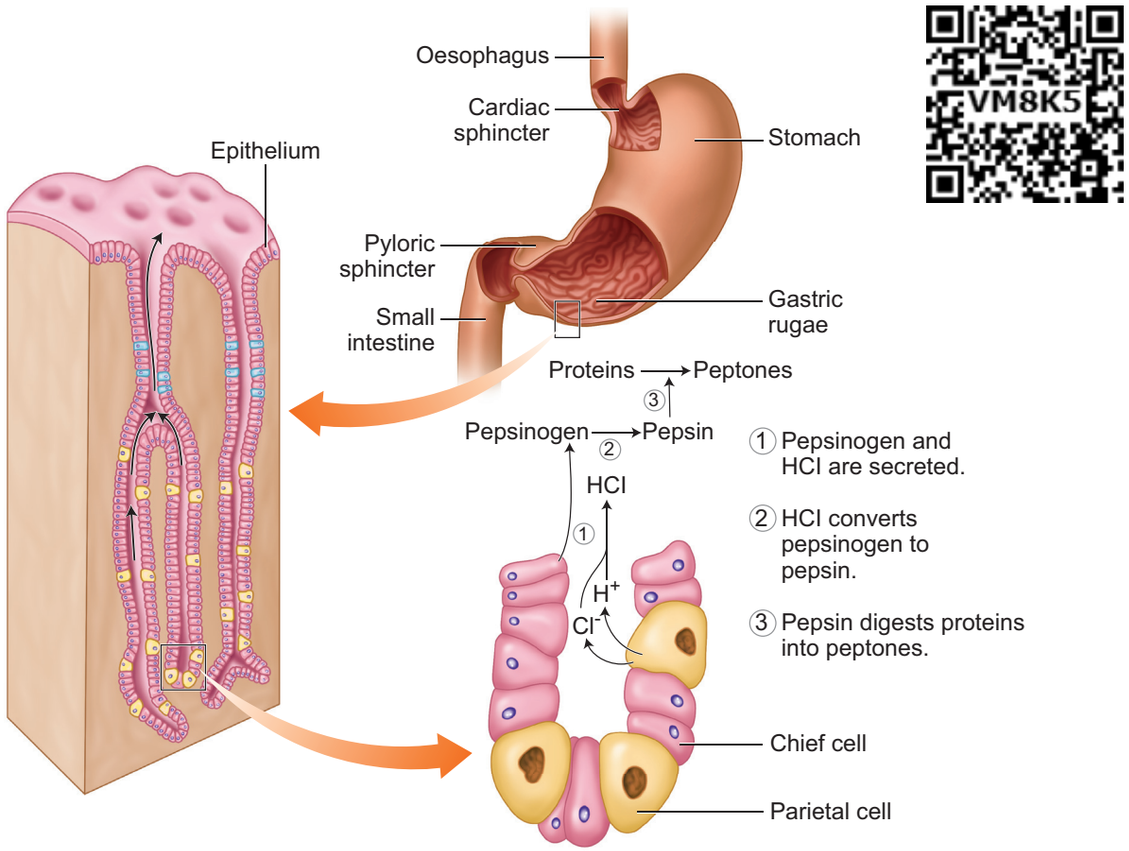


Figure.5.8 The stomach and gastric secretions



Bicarbonates in the saliva make the pH 5.4 to 7.4. If the bicarbonates level in saliva is reduced the saliva becomes acidic and the tooth enamel may get dissolved.

What would happen if HCl is not secreted in the stomach?

Digestion in the small intestine

The bile, pancreatic juice and intestinal juice are the secretions released into the small intestine. Movements generated by the muscularis layer of the small intestine helps in the thorough mixing of the food with various secretions in the intestine and thereby facilitate digestion.

The bile contains bile pigments (bilirubin and biliverdin) as the break down products of hemoglobin of dead RBCs, bile salts, cholesterol and phospholipids but has no enzymes. Bile helps in emulsification of fats. Bile salts reduce the surface tension of fat droplets and break them into small globules. Bile also activates lipases to digest lipids Proteins and partially digested proteins in the chyme on reaching the intestine are acted upon by the proteolytic enzymes of pancreatic juice.The pancreatic juice contains enzymes such as trypsinogen, chymotrypsinogen, carboxypeptidases, pancreatic amylases, pancreatic lipases and nucleases. Trypsinogen is activated by an enzyme, enterokinase, secreted by the intestinal mucosa into active trypsin, which in turn activates the enzyme chymotrypsinogen in the pancreatic juice.

XI Std Zoology Chapter-5 EM.indd 80 30/12/2022 10:38:46